

ECA56 –Advanced WirelessCommunication Systems / 5G / 6GCommunication Systems

Ex.No:1: Simulation of FDMA in MATLAB Date

Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users and determine the channel bandwidth assigned to each user.

CODE :

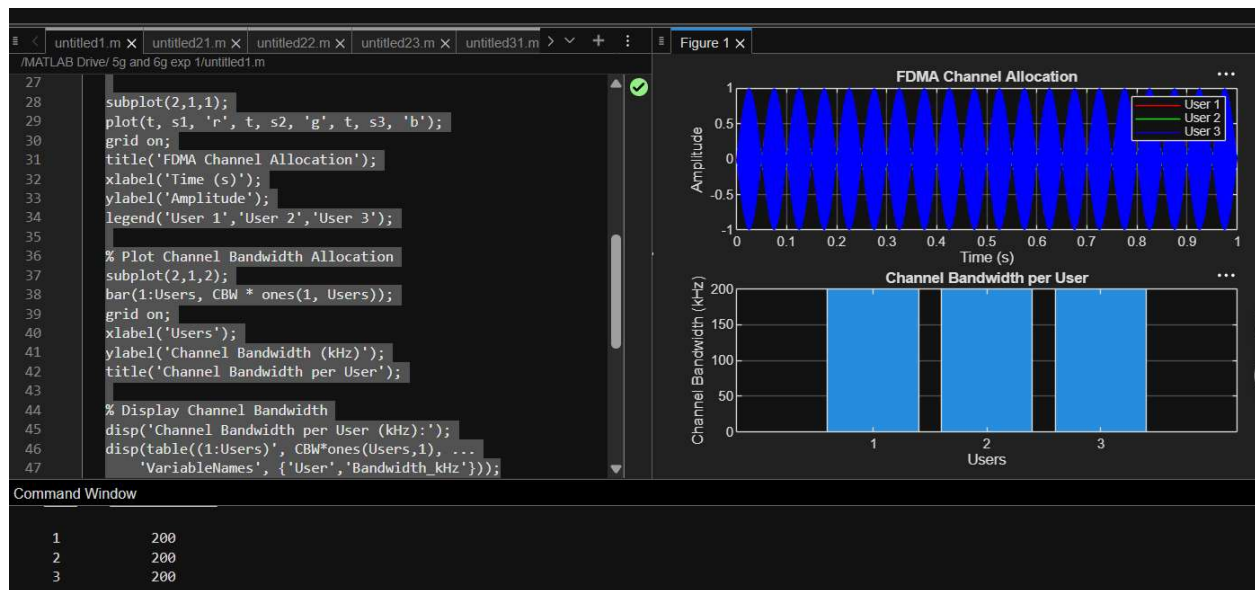
```
clc;
clear;
close all;
% Sampling parameters
fs = 1000;
t = 0:1/fs:1;
% FDMA Parameters
TBW = 600; % Total Bandwidth (kHz)
Users = 3; % Number of Users
CBW = TBW / Users; % Channel Bandwidth per User
% Carrier Frequencies (Hz)
fc = [100 300 500];
% Message Signal
m = sin(2*pi*10*t);
% Modulated Signals
s1 = m .* cos(2*pi*fc(1)*t);
s2 = m .* cos(2*pi*fc(2)*t);
s3 = m .* cos(2*pi*fc(3)*t);
% Plot FDMA Signals
figure;
subplot(2,1,1);
plot(t, s1, 'r', t, s2, 'g', t, s3, 'b');
grid on;
title('FDMA Channel Allocation');
xlabel('Time (s)');
ylabel('Amplitude');
legend('User 1', 'User 2', 'User 3');
% Plot Channel Bandwidth Allocation
subplot(2,1,2);
bar(1:Users, CBW * ones(1, Users));
grid on;
```

```

xlabel('Users');
ylabel('Channel Bandwidth (kHz)');
title('Channel Bandwidth per User');
% Display Channel Bandwidth
disp('Channel Bandwidth per User (kHz):');
disp(table((1:Users)', CBW*ones(Users,1), ...
'VariableNames', {'User','Bandwidth_kHz'}));

```

Output :



Objective 2:

To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth.

Output Parameter: Data Rate (kbps)

CODE :

```

clc;
clear;
close all;
% FDMA Data Rate Calculation
BW = 200; % Channel Bandwidth (kHz)
Users = 3; % Number of Users
M = 2; % BPSK Modulation
% Data Rate Calculation
DataRate = BW * log2(M); % Data Rate (kbps)
Rate = DataRate * ones(1, Users);
% Display Output

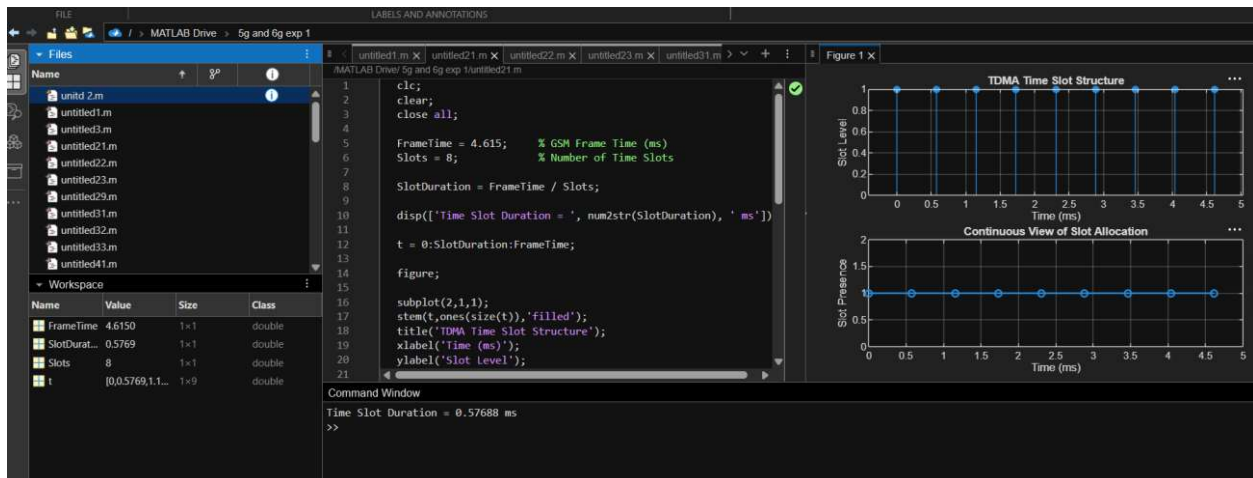
```

```

disp('User Data Rate (kbps)');
fprintf('-----\n');
fprintf('User\tData Rate (kbps)\n');
for i = 1:Users
fprintf('%d\t\t%.0f\n', i, Rate(i));
end
% Plot
figure;
bar(1:Users, Rate);
xlabel('Users');
ylabel('Data Rate (kbps)');
title('FDMA Data Rate for Each User');
xticks(1:Users);
grid on;

```

Output :



Objective 3:

To evaluate the communication quality of the FDMA system by calculating the Signal-to-Noise Ratio (SNR) for different users.

Output Parameter: Signal-to-Noise Ratio (SNR) (dB)

CODE :

```

clc;
clear;
close all;
% FDMA SNR Calculation
Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)

```

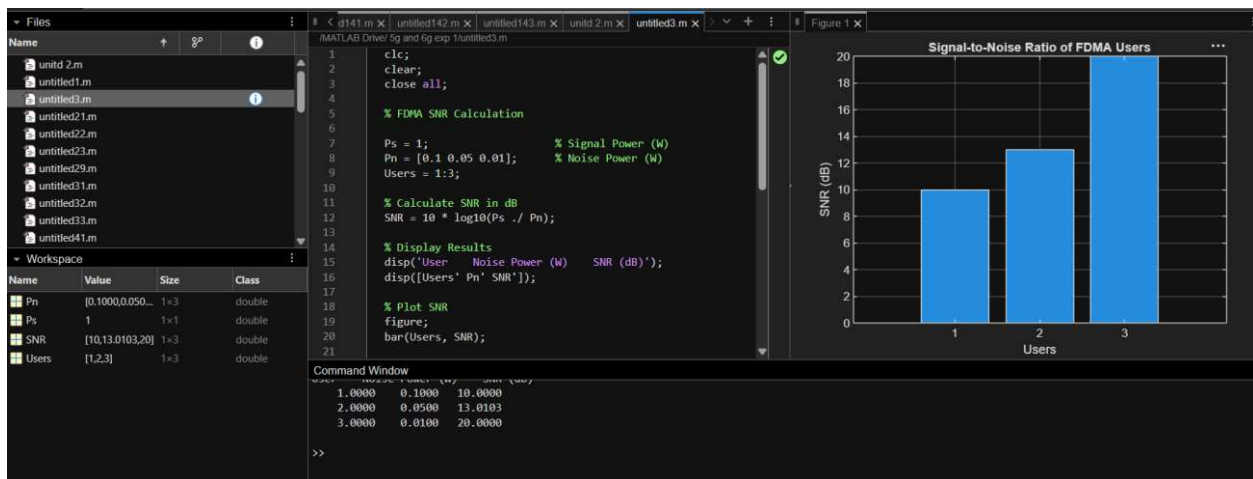
```

Users = 1:3;
% Calculate SNR in dB
SNR = 10 * log10(Ps ./ Pn);
% Display Results
disp('User Noise Power (W) SNR (dB)');
disp([Users' Pn' SNR']);
% Plot SNR
figure;

bar(Users, SNR);
xlabel('Users');
ylabel('SNR (dB)');
title('Signal-to-Noise Ratio of FDMA Users');
xticks(Users);
grid on;

```

Output:



Ex.No. 2: Implementing TDMA Slot Allocation in GSM. Date

Objective:

1. To calculate the Time Slot Duration in a GSM TDMA frame by dividing the total frame time into equal time slots. This helps in understanding the time structure of GSM communication.

Program :

```

clc;
clear;
close all;
FrameTime = 4.615; % GSM Frame Time (ms)
Slots = 8; % Number of Time Slots

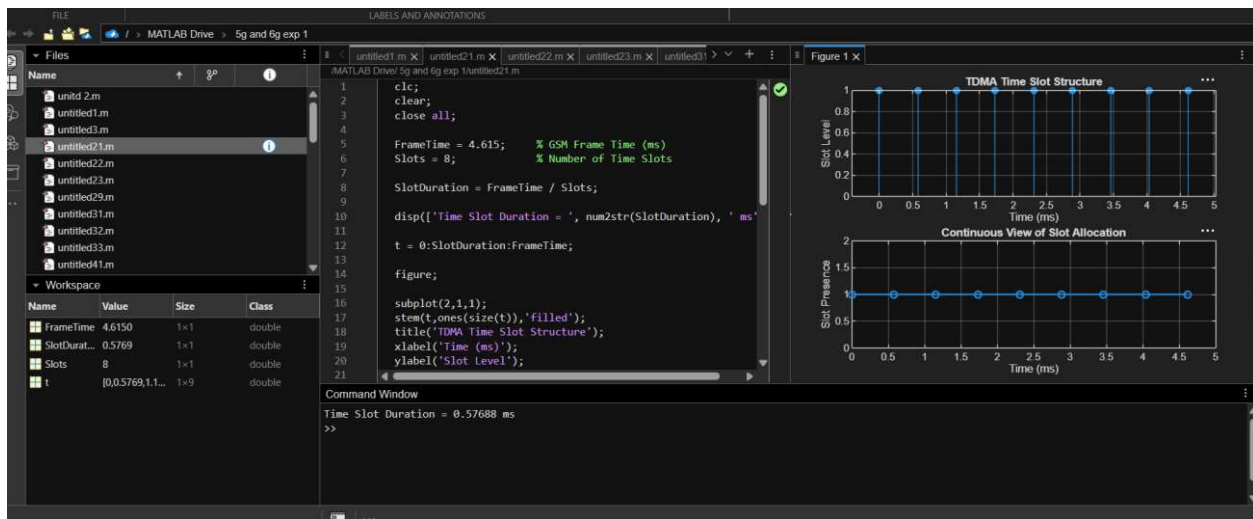
```

```

SlotDuration = FrameTime / Slots;
disp(['Time Slot Duration = ', num2str(SlotDuration), ' ms']);
t = 0:SlotDuration:FrameTime;
figure;
subplot(2,1,1);
stem(t,ones(size(t)),'filled');
title('TDMA Time Slot Structure');
xlabel('Time (ms)');
ylabel('Slot Level');
grid on;
subplot(2,1,2);
plot(t,ones(size(t)),'-o','LineWidth',1.5);
title('Continuous View of Slot Allocation');
xlabel('Time (ms)');
ylabel('Slot Presence');
grid on;

```

Output :



Objective:2

To determine the Frame Utilization (%) based on the number of allocated time slots compared to total available slots. This evaluates the efficiency of TDMA resource usage.

Program:

```

clc;
clear;
close all;
TotalSlots = 8;

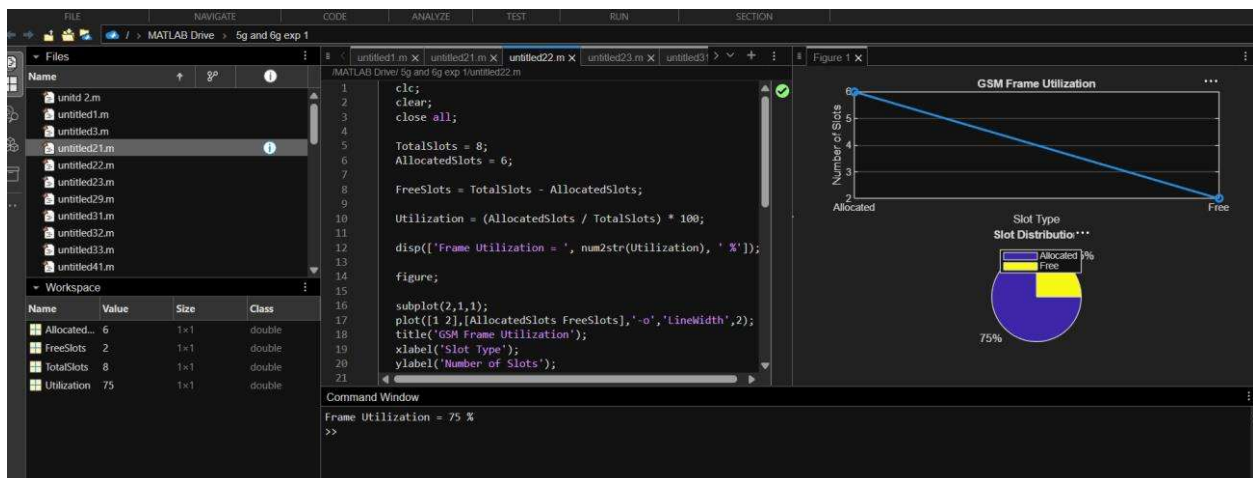
```

```

AllocatedSlots = 6;
FreeSlots = TotalSlots - AllocatedSlots;
Utilization = (AllocatedSlots / TotalSlots) * 100;
disp(['Frame Utilization = ', num2str(Utilization), ' %']);
figure;
subplot(2,1,1);
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2);
title('GSM Frame Utilization');
xlabel('Slot Type');
ylabel('Number of Slots');
xticks([1 2]);
xticklabels({'Allocated','Free'});
grid on;
subplot(2,1,2);
pie([AllocatedSlots FreeSlots]);
title('Slot Distribution');
legend('Allocated','Free');

```

Output :



Objective 3:

To compute the Number of Allocated Time Slots for multiple users in the GSM system. This ensures proper scheduling and interference-free communication.

```

Program : clc;
clear;
close all;
Users = 1:8; % Total Users
TotalSlots = 5; % Available Slots
Alloc = zeros(1,8);
Alloc(1:TotalSlots) = 1;
Blocked = 1 - Alloc;

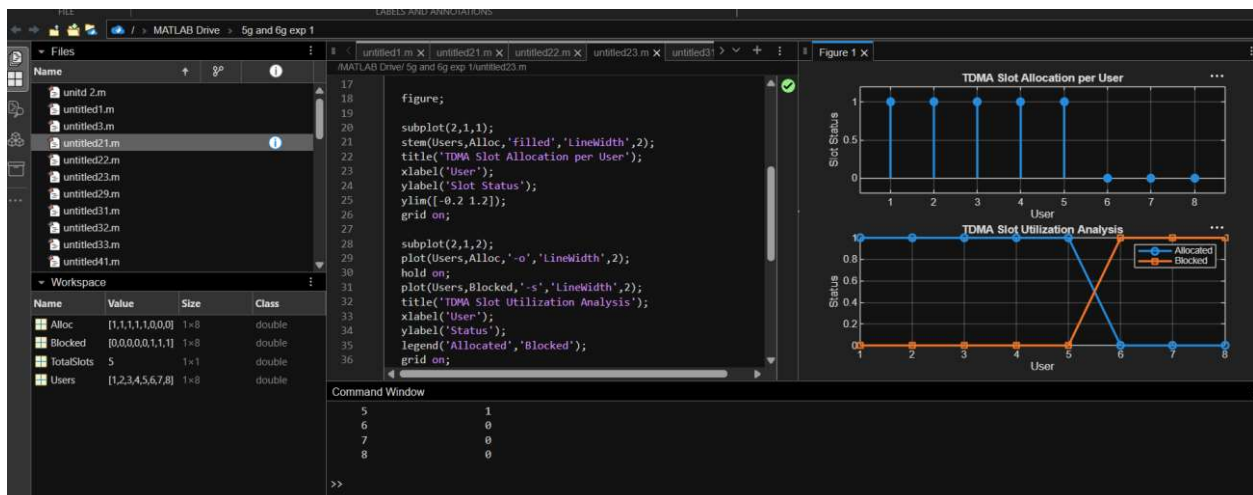
```

```

disp('User Wise Slot Allocation');
disp(table(Users',Alloc', ...
'VariableNames',{'User','Status_1_Allocated_0_Blocked'}));
figure;
subplot(2,1,1);
stem(Users,Alloc,'filled','LineWidth',2);
title('TDMA Slot Allocation per User');
xlabel('User');
ylabel('Slot Status');
ylim([-0.2 1.2]);
grid on;
subplot(2,1,2);
plot(Users,Alloc,'-o','LineWidth',2);
hold on;
plot(Users,Blocked,'-s','LineWidth',2);
title('TDMA Slot Utilization Analysis');
xlabel('User');
ylabel('Status');
legend('Allocated','Blocked');
grid on;

```

Output :



Ex.N.3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation Loss. Date

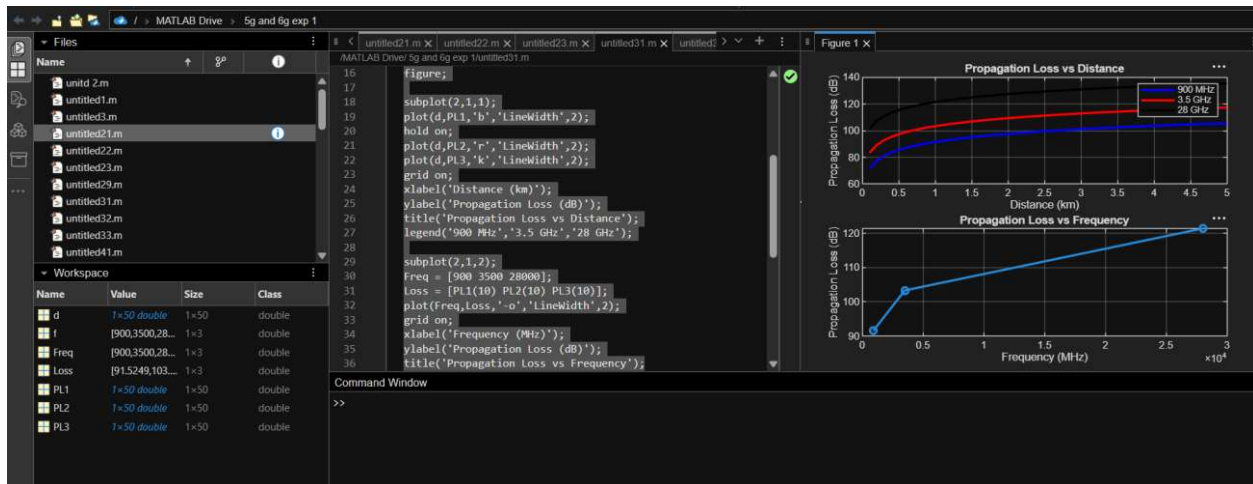
Objective 1:

To compute the Propagation Loss (dB) for Sub-1 GHz, Mid-band, and mmWave frequency ranges using standard path loss models

Program:

```
clc;
clear;
close all;
% Distance (km)
d = 0.1:0.1:5;
% Frequency (MHz)
f = [900 3500 28000];
% Free Space Path Loss
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure;
subplot(2,1,1);
plot(d,PL1,'b','LineWidth',2);
hold on;
plot(d,PL2,'r','LineWidth',2);
plot(d,PL3,'k','LineWidth',2);
grid on;
xlabel('Distance (km)');
ylabel('Propagation Loss (dB)');
title('Propagation Loss vs Distance');
legend('900 MHz','3.5 GHz','28 GHz');
subplot(2,1,2);
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq,Loss,'-o','LineWidth',2);
grid on;
xlabel('Frequency (MHz)');
ylabel('Propagation Loss (dB)');
title('Propagation Loss vs Frequency');
```

Output:



Objective 2:

To evaluate the impact of different Operating Frequencies (GHz) on signal attenuation and coverage performance.

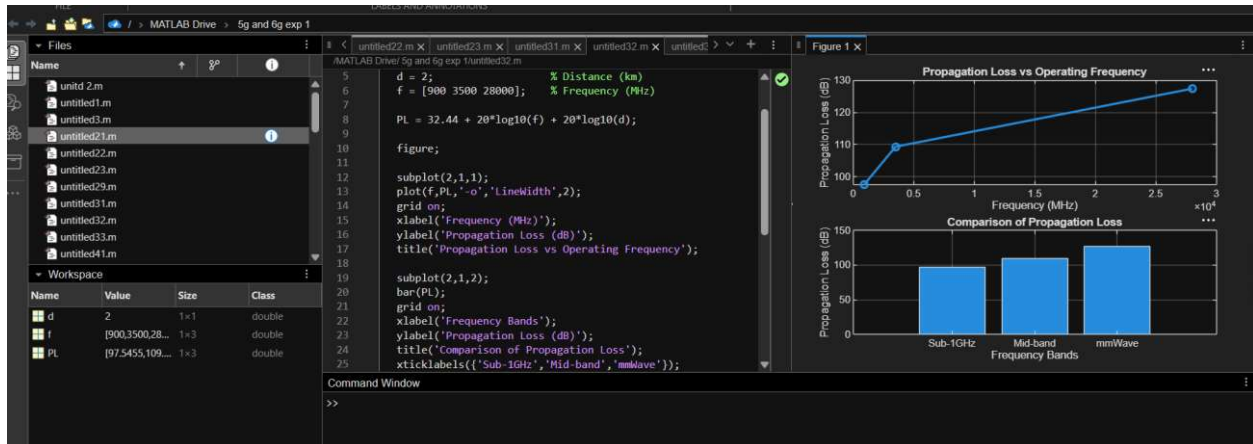
Program :

```

clc;
clear;
close all;
d = 2; % Distance (km)
f = [900 3500 28000]; % Frequency (MHz)
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure;
subplot(2,1,1);
plot(f, PL, '-o', 'LineWidth', 2);
grid on;
xlabel('Frequency (MHz)');
ylabel('Propagation Loss (dB)');
title('Propagation Loss vs Operating Frequency');
subplot(2,1,2);
bar(PL);
grid on;
xlabel('Frequency Bands');
ylabel('Propagation Loss (dB)');
title('Comparison of Propagation Loss');
xticklabels({'Sub-1GHz', 'Mid-band', 'mmWave'});

```

Output:

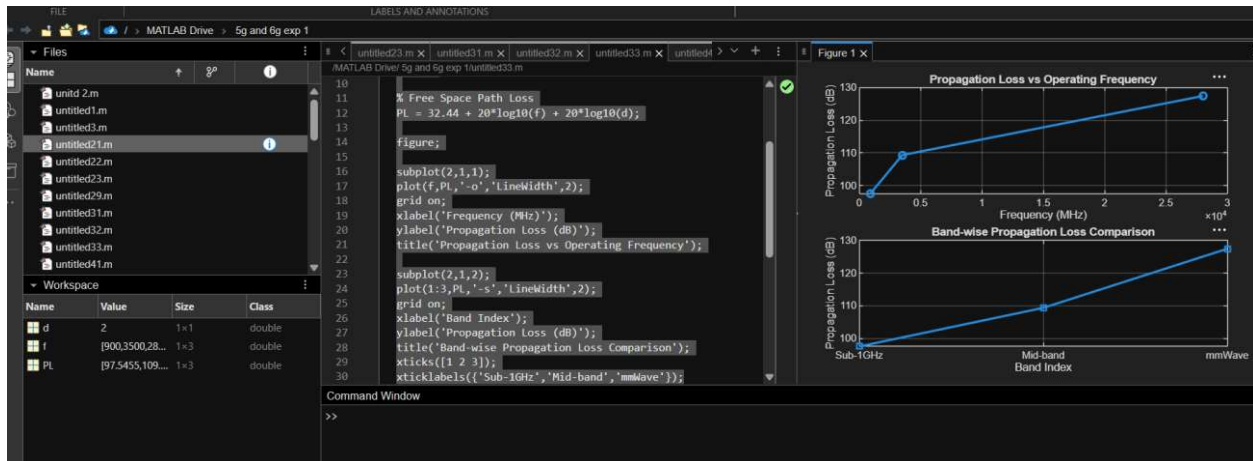


Objective 3: To analyze how Communication Distance (km) affects propagation loss across different spectrum bands.

Program :

```
clc;
clear;
close all;
% Frequency bands (MHz)
f = [900 3500 28000];
% Distance (km)
d = 2;
% Free Space Path Loss
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure;
subplot(2,1,1);
plot(f,PL, '-o', 'LineWidth',2);
grid on;
xlabel('Frequency (MHz)');
ylabel('Propagation Loss (dB)');
title('Propagation Loss vs Operating Frequency');
subplot(2,1,2);
plot(1:3,PL, '-s', 'LineWidth',2);
grid on;
xlabel('Band Index');
ylabel('Propagation Loss (dB)');
title('Band-wise Propagation Loss Comparison');
xticks([1 2 3]);
xticklabels({'Sub-1GHz', 'Mid-band', 'mmWave'});
```

Output :



Ex.No.4: OFDMA Resource Block Allocation Date

Objective 1:

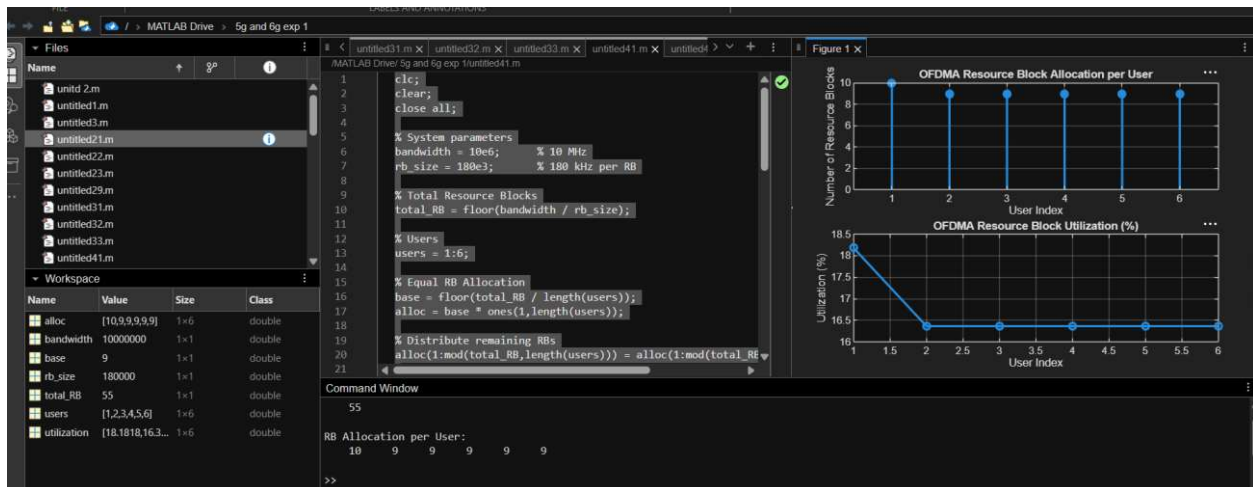
```

clc;
clear;
close all;
% System parameters
bandwidth = 10e6; % 10 MHz
rb_size = 180e3; % 180 kHz per RB
% Total Resource Blocks
total_RB = floor(bandwidth / rb_size);
% Users
users = 1:6;
% Equal RB Allocation
base = floor(total_RB / length(users));
alloc = base * ones(1,length(users));
% Distribute remaining RBs
alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) + 1;
utilization = (alloc / total_RB) * 100;
disp('Total Resource Blocks:');
disp(total_RB);
disp('RB Allocation per User:');
disp(alloc);
figure;
subplot(2,1,1);
stem(users,alloc,'filled','LineWidth',2);
title('OFDMA Resource Block Allocation per User');
xlabel('User Index');
ylabel('Number of Resource Blocks');
grid on;
subplot(2,1,2);
plot(users,utilization,'-o','LineWidth',2);
title('OFDMA Resource Block Utilization (%)');

```

```
xlabel('User Index');
ylabel('Utilization (%)');
grid on;
```

Output :



Objective 2:

To evaluate the Resource Block Utilization (%) by analyzing how efficiently available RBs are assigned to active users.

Program :

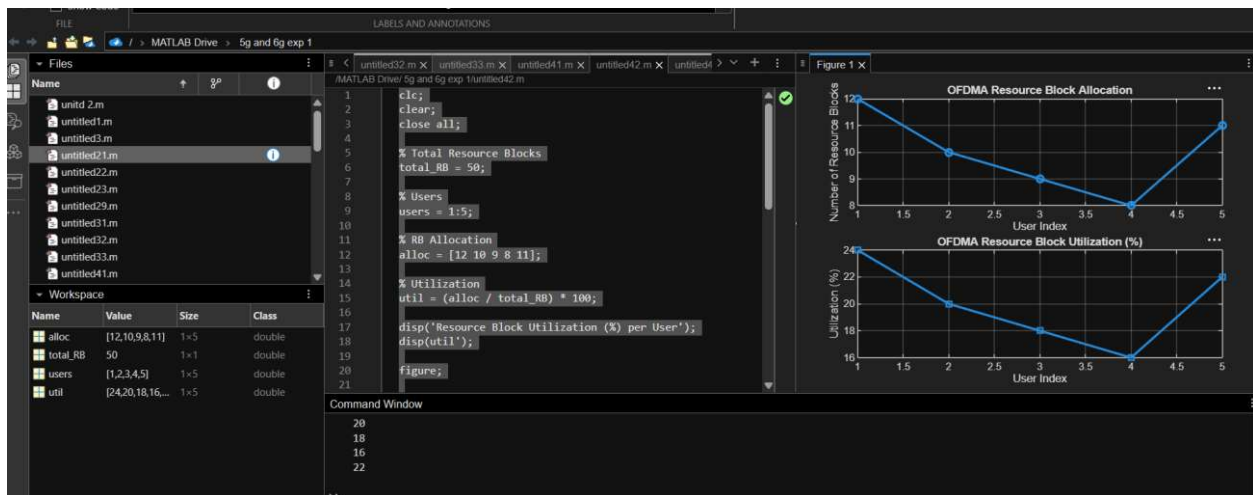
```
clc;
clear;
close all;
% Total Resource Blocks
total_RB = 50;
% Users
users = 1:5;
% RB Allocation
alloc = [12 10 9 8 11];
% Utilization
util = (alloc / total_RB) * 100;
disp('Resource Block Utilization (%) per User');
disp(util);
figure;
subplot(2,1,1);
plot(users,alloc,'-o','LineWidth',2);
title('OFDMA Resource Block Allocation');
```

```

xlabel('User Index');
ylabel('Number of Resource Blocks');
grid on;
subplot(2,1,2);
plot(users,util,'-s','LineWidth',2);
title('OFDMA Resource Block Utilization (%)');
xlabel('User Index');
ylabel('Utilization (%)');
grid on;

```

Output:



Objective3:

To calculate the User Throughput (Mbps) based on allocated resource blocks and Modulation efficiency in the OFDMA system.

Program:

```

clc;
clear;
close all;
% Bandwidth per Resource Block
rb_bw = 180e3; % 180 kHz
% Users
users = 1:5;
% Allocated Resource Blocks
alloc_RB = [12 10 9 8 11];
% Modulation Efficiency
mod_eff = [2 4 6 4 2];
% Throughput Calculation
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

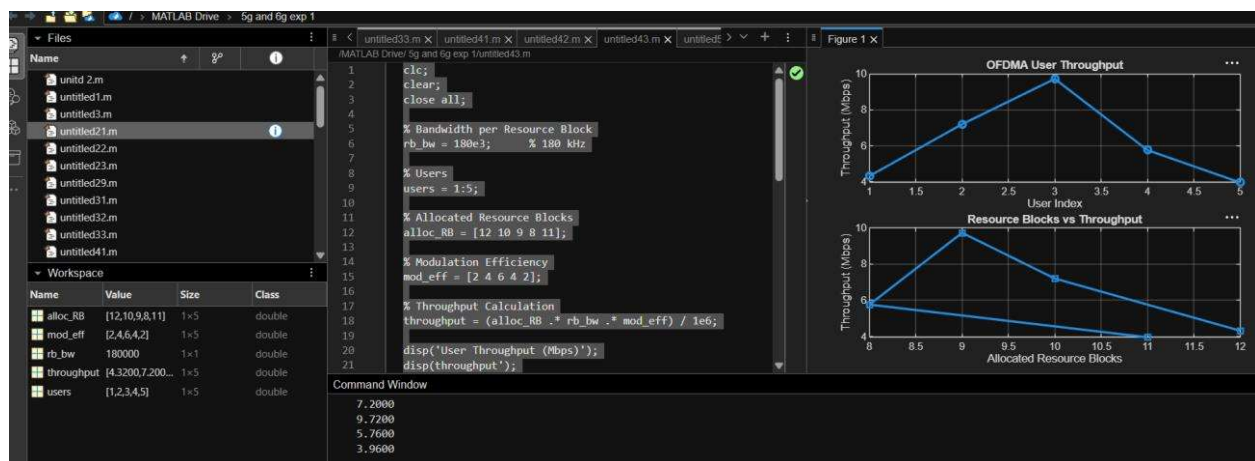
```

```

disp('User Throughput (Mbps)');
disp(throughput);
figure;
subplot(2,1,1);
plot(users,throughput, '-o', 'LineWidth',2);
title('OFDMA User Throughput');
xlabel('User Index');
ylabel('Throughput (Mbps)');
grid on;
subplot(2,1,2);
plot(alloc_RB,throughput, '-s', 'LineWidth',2);
title('Resource Blocks vs Throughput');
xlabel('Allocated Resource Blocks');
ylabel('Throughput (Mbps)');
grid on;

```

Output:



Ex.No:5: MIMO System Simulation with BER Analysis Date

Objective 1 :

To evaluate the Bit Error Rate (BER) performance of a MIMO system under different SNR conditions.

Program:

```

clc; clear; close all;
% SNR range
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters

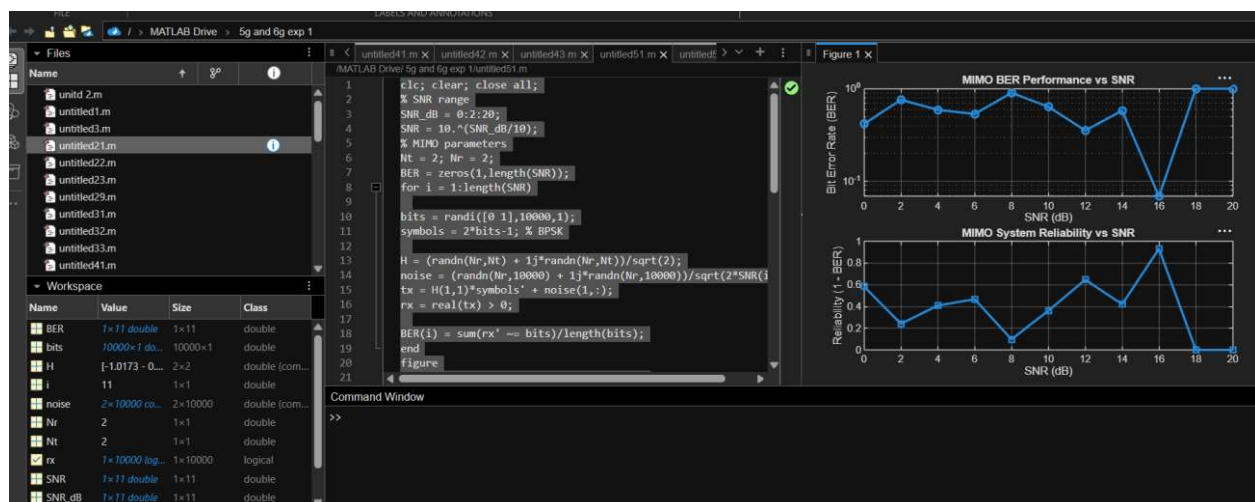
```

```

Nt = 2; Nr = 2;
BER = zeros(1,length(SNR));
for i = 1:length(SNR)
bits = randi([0 1],10000,1);
symbols = 2*bits-1; % BPSK
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
tx = H(1,1)*symbols' + noise(1,:);
rx = real(tx) > 0;
BER(i) = sum(rx' ~= bits)/length(bits);
end
figure
% ----- Graph 1: BER vs SNR -----
subplot(2,1,1)
semilogy(SNR_dB,BER,'-o','LineWidth',2)
title('MIMO BER Performance vs SNR')
xlabel('SNR (dB)')
ylabel('Bit Error Rate (BER)')
grid on
% ----- Graph 2: SNR vs Quality Indicator -----
subplot(2,1,2)
plot(SNR_dB,1-BER,'-s','LineWidth',2)
title('MIMO System Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (1 - BER)')
grid on

```

Output :



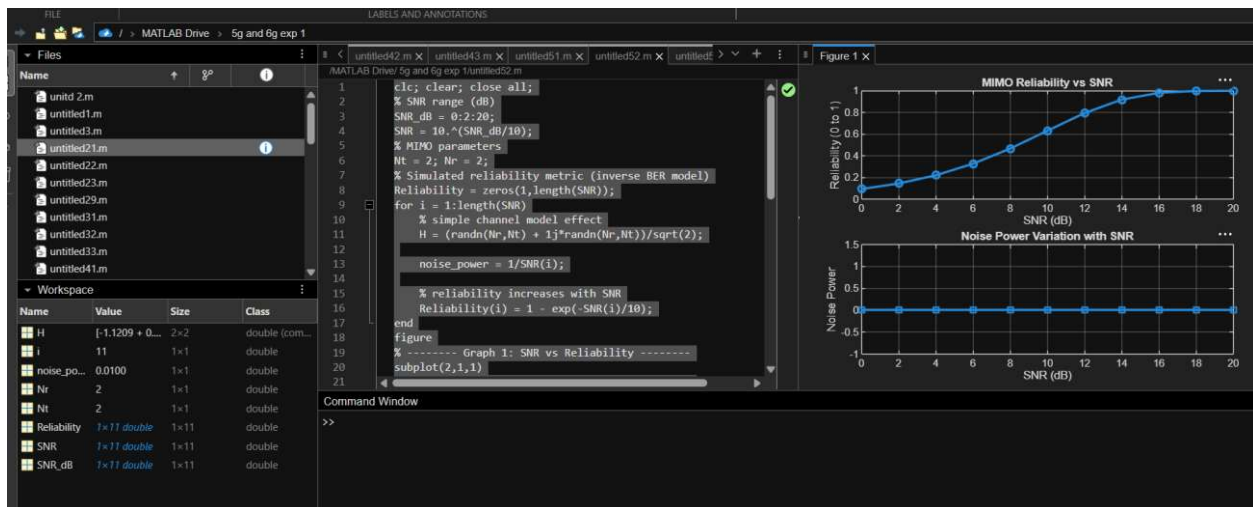
Objective 2:

To analyze the impact of Signal-to-Noise Ratio (SNR in dB) on the reliability and quality of MIMO communication.

Program :

```
clc; clear; close all;
% SNR range (dB)
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2; Nr = 2;
% Simulated reliability metric (inverse BER model)
Reliability = zeros(1,length(SNR));
for i = 1:length(SNR)
% simple channel model effect
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
noise_power = 1/SNR(i);
% reliability increases with SNR
Reliability(i) = 1 - exp(-SNR(i)/10);
end
figure
% ----- Graph 1: SNR vs Reliability -----
subplot(2,1,1)
plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
title('MIMO Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (0 to 1)')
grid on
% ----- Graph 2: SNR vs Noise Effect -----
subplot(2,1,2)
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)
title('Noise Power Variation with SNR')
xlabel('SNR (dB)')
ylabel('Noise Power')
grid on
```

Output:



Objective 3:

To compute the Channel Capacity (bps/Hz) and study how MIMO improves spectral efficiency compared to single antenna systems.

Program :

```
clc; clear; close all;
% Resource Block parameters
rb_bw = 180e3; % 180 kHz per RB
% Users
users = 1:5;
% Allocated Resource Blocks per user
alloc_RB = [10 12 8 15 5];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
% Display results
disp('User Throughput (Mbps):')
disp(throughput)
figure
% ----- Graph 1: Throughput per User -----
subplot(2,1,1)
plot(users, throughput, '-o','LineWidth',2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2: RB vs Throughput -----
subplot(2,1,2)
plot(alloc_RB, throughput, '-s','LineWidth',2)
```

```

title('Resource Blocks vs Throughput')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on

```

Output:

